MARKING SCHEME

Model Paper — 05

Mathematics II

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Note for teachers:

- Candidates answer **five** questions from Part A (Questions 1 – 6) and **five** questions from Part B (Questions 7 – 12). Each question carries 10 marks; total 100 marks.
- The marks for each step are shown in the right-hand column and add up to the marks printed for each part of the question.
- Award full marks for any correct alternative method. For graphs, constructions, measured values and answers read from tables, accept a reasonable range.

Part A — Answers

Question 1

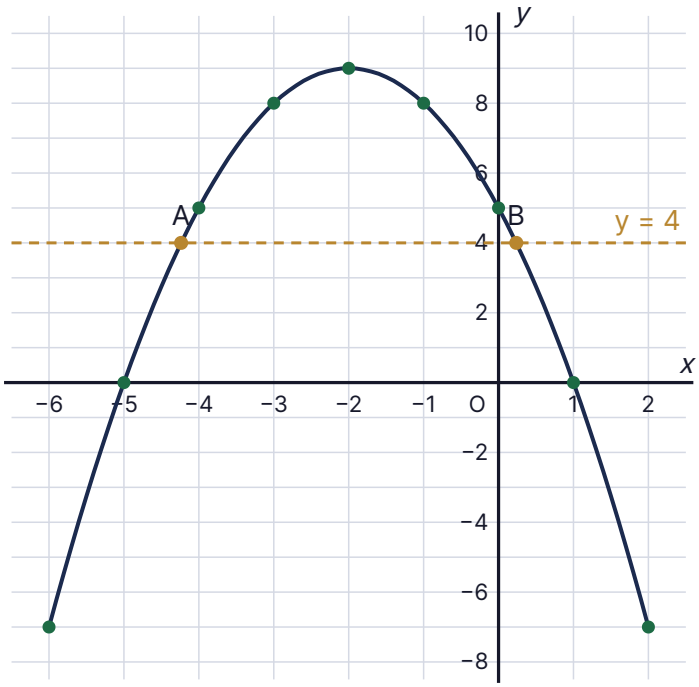
(i)	
Rate = $\frac{10\,140}{78\,000} \times 100$	**1**
= 13% per annum	**1**
(ii)	
1st month: Rs. 78 000; 2nd month: 78 000 − 6 500 = Rs. 71 500	**1**
12th month: 78 000 − 11 × 6 500 = Rs. 6 500	**1**
(iii)	
Sum = $\frac{12}{2}$ (78 000 + 6 500) (AP: a = 78 000, l = 6 500, n = 12)	**1**
= 6 × 84 500 = 507 000	**1**
(iv)	
Total interest = $\frac{507\,000 \times r}{100 \times 12}$	**1**
$\frac{507\,000 \times r}{1\,200}$ = 10 140	**1**
$r = \frac{10\,140 \times 1\,200}{507\,000}$ = 24	**1**
(v)	
Ruwan's rate 24% per annum is greater than Nimesha's rate 13% per annum.	**1**

Total marks: 10

Question 2

(i)	
The axis of symmetry is $x = -2$ (the points $x = -4$ and $x = 0$ are symmetric about it), so $y = 5$ when $x = 0$	**1**
(ii)	
Suitable scales on both axes (e.g. 2 cm = 1 unit on x, 1 cm = 1 unit on y) and the axes drawn	**1**
Points plotted correctly: (−6, −7), (−5, 0), (−4, 5), (−3, 8), (−2, 9), (−1, 8), (0, 5), (1, 0), (2, −7) (at least 7 correct)	**1**
A smooth curve (an inverted parabola with maximum point (−2, 9)) through the points	**1**
(iii)	
(a) $x = -5$ and $x = 1$	**1**
(b) $y = -(x - 1)(x + 5)$ ($p = -1$, $q = 5$; check: at $x = 0$, $y = 5$)	**1**
(iv)	
(a) Line cuts the graph at $x \approx -4.2$ and $x \approx 0.2$ (accept ±0.1)	**1**
Range: $-4.2 < x < 0.2$	**1**
(b) AB = 0.2 − (−4.2) (difference of the two x values read from the graph)	**1**
AB ≈ 4.5 cm (accept 4.4 – 4.5; exact value $2\sqrt{5} \approx 4.47$)	**1**

Total marks: 10



The graph with the line y = 4 (points A and B)

Question 3

(i)	
In the right-angled triangle ABC (right angle at B): BC = AC × sin 33°30′	**1**
= 6 × 0.5519 (from the sine table)	**1**
= 3.311 m (≈ 3.31 m)	**1**
(ii)	
BD = 2 × BC (C is the mid-point of BD)	**1**
= 2 × 3.311 = 6.623 m	**1**
$\tan \angle DEB = \frac{BD}{BE} = \frac{6.623}{7.2} = 0.9198$ (≈ 0.92)	**1**
(iii)	
From the tangent table: tan 42°36′ = 0.9195, so ∠DEB ≈ 42°36′ (accept 42°36′ – 42°42′)	**1**
Difference = ∠DEB − ∠CAB = 42°36′ − 33°30′	**1**
= 9°6′	**1**
9°6′ > 8°, so ∠DEB exceeds ∠CAB by more than 8°	**1**

Total marks: 10

Question 4

(i)	
Area of the triangle = $\frac{1}{2} \times 2x \times 2x = 2x^2$	**1**
Area of the bed = $(x + 1)(x - 3) = x^2 - 2x - 3$	**1**
$2x^2 - (x^2 - 2x - 3) = 100 \Rightarrow x^2 + 2x - 97 = 0$	**1**
(ii)	
$x = \frac{-2 \pm \sqrt{(2^2 - 4 \times 1 \times (-97))}}{2}$	**1**
$= \frac{-2 \pm \sqrt{392}}{2}$	**1**
$\sqrt{392} = 14\sqrt{2}$, so $x = -1 \pm 7\sqrt{2}$	**1**
x must be positive: $x = -1 + 7 \times 1.41 = 8.87 \approx 9$ m	**1**
(iii)	
Longest side (hypotenuse) = $\sqrt{((2x)^2 + (2x)^2)} = 2\sqrt{2}\,x$	**1**
$= 2\sqrt{2}\,(7\sqrt{2} - 1) = 28 - 2\sqrt{2}$	**1**
$= 28 - 2 \times 1.41 = 25.18$ m > 25 m	**1**

Total marks: 10

Question 5

(i)	
$x + y = 36$	**1**
$65x + 110y = 3\,060$	**1**
Substituting $y = 36 - x$: $65x + 3\,960 - 110x = 3\,060$	**1**
$45x = 900$, $x = 20$	**1**
$y = 16$: 20 small pots and 16 large pots	**1**
(ii)	
Let a small pot hold m plants; a large pot holds $m + 3$	**1**
$20m + 16(m + 3) = 336$	**2**
$36m + 48 = 336$, $36m = 288$	**1**
$m = 8$ plants	**1**

Total marks: 10

Question 6

(i)	
Class mid-values 64.5, 74.5, 84.5, 94.5, 104.5, 114.5	**1**
$\Sigma fx = 3(64.5) + 7(74.5) + 9(84.5) + 9(94.5) + 6(104.5) + 6(114.5) = 3\,640$	**1**
Mean $= \frac{3\,640}{40} = 91$ runs	**1**
(ii)	
Expected total $= 12 \times 91$	**1**
$= 1\,092$ runs	**1**
(iii)	
Sponsor's payment $= 1\,092 \times 25 =$ Rs. 27 300	**1**
Ground charges $= 12 \times 1\,200 =$ Rs. 14 400	**1**
Net income $= 27\,300 - 14\,400 =$ Rs. 12 900	**1**
(iv)	
Matches with fewer than 80 runs are in the classes 60 – 69 and 70 – 79: $3 + 7 = 10$ matches	**1**
Minimum total $= 3 \times 60 + 7 \times 70 = 670$ runs	**1**

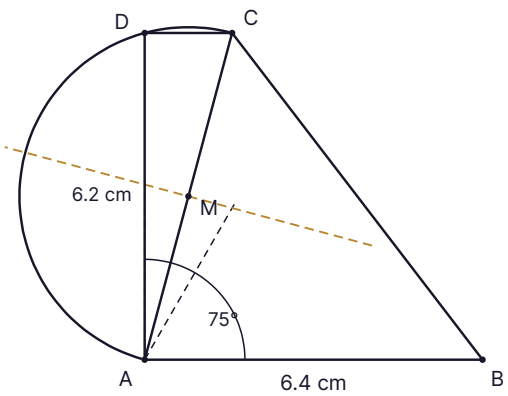
Total marks: 10

Part B — Answers

Question 7

(i)	
AB = 6.4 cm drawn (accept 6.3 – 6.5 cm)	**1**
60° constructed at A (equilateral-triangle arcs visible)	**1**
90° constructed at A (perpendicular construction arcs visible)	**1**
Bisector of the angle between the 60° and 90° lines: ∠BAX = 75° (accept 74° – 76°)	**1**
(ii)	
C marked on AX with AC = 6.4 cm (compass arc from A), BC joined; the triangle ABC has ∠BAC = 75°, ∠ABC = ∠ACB = 52.5°	**1**
(iii)	
Perpendicular bisector of AC drawn with correct arcs, giving the mid-point M of AC	**1**
Semicircle with centre M and radius MA (3.2 cm) drawn on the side of AC away from B	**1**
(iv)	
Line through C parallel to AB constructed (equal angles / alternate angles with arcs visible)	**1**
D marked where this line meets the semicircle, and AD joined (AD ⊥ AB, AD ≈ 6.2 cm; accept 6.1 – 6.3 cm)	**1**
(v)	
∠ACD = ∠CAB = 75° (alternate angles, CD ∥ AB) and ∠ADC = 90° (angle in a semicircle), so ∠CAD = 180° − 90° − 75° = 15°	**1**

Total marks: 10



Expected construction (approximate, not to scale)

Question 8

(i)	
In △AGE and △DCE: ∠AEG = ∠DEC (vertically opposite angles)	**1**
∠GAE = ∠CDE (alternate angles, AG ∥ DC)	**1**
AE = ED (E is the mid-point of AD), so △AGE ≅ △DCE (angle, side, angle)	**1**
(ii)	
From the congruence, GE = EC; also AE = ED	**1**
The diagonals AD and GC of AGDC bisect each other, so AGDC is a parallelogram	**1**
(iii)	
In △AFH and △BFC: ∠AFH = ∠BFC (vertically opposite angles)	**1**
∠FAH = ∠FBC (alternate angles, AH ∥ BC)	**1**
Hence the third angles are equal (∠AHF = ∠BCF), so the triangles are equiangular	**1**
(iv)	
$\frac{AH}{BC} = \frac{AF}{FB}$ (corresponding sides of similar triangles)	**1**
AF = $\frac{1}{3}$ AB gives AF : FB = 1 : 2, so AH = $\frac{1}{2}$ BC = $\frac{1}{2}$ AD (BC = AD)	**1**

Total marks: 10

Question 9

(i)	
Volume of one pyramid = $\frac{1}{3} \times a^2 \times 4a = \frac{4a^3}{3}$	**1**
Volume of 6 pyramids = $6 \times \frac{4a^3}{3} = 8a^3$	**1**
Volume of the cylinder = $\pi r^2 \times 5r = 5\pi r^3$	**1**
$8a^3 = 5\pi r^3 \Rightarrow a^3 = \frac{5\pi}{8} r^3$	**1**
(ii)	
$a^3 = \frac{5 \times 3.14 \times 5.2^3}{8}$; lg a^3 = lg 5 + lg 3.14 + 3 lg 5.2 − lg 8 = 0.6990 + 0.4969 + 3 × 0.7160 − 0.9031	**1**
= 2.4408, so a^3 = antilog 2.4408 ≈ 275.9 (accept 275 – 276)	**1**
lg a = 2.4408 ÷ 3 = 0.8136, so a = 6.51 ≈ 6.5 cm	**1**
(iii)	
Volume of the cuboid = $a^2 \times 4a = 4a^3$; volume of the pyramid cut out = $\frac{4a^3}{3}$	**1**
Remaining volume = $4a^3 - \frac{4a^3}{3} = \frac{8a^3}{3}$	**1**
= $\frac{8}{3} \times 275.9 \approx 735.7$ cm^3 (accept 734 – 737)	**1**

Total marks: 10

Question 10

(i)	
20, 25, 30	**1**
(ii)	
$T_{14} = a + 13d = 20 + 13 \times 5$	**1**
= 85 balls	**1**
(iii)	
$S_{14} = \frac{14}{2}(20 + 85)$	**1**
$= 7 \times 105 = 735$ balls	**1**
(iv)	
Ruwan: $n = 15$, $d = 3$, $S_{15} = 735$: $\frac{15}{2}(2a + 14 \times 3) = 735$	**1**
$2a + 42 = 98$	**1**
$a = 28$ balls	**1**
(v)	
$20 + 5(n - 1) = 28 + 3(n - 1)$	**1**
$2(n - 1) = 8$, $n = 5$: on the 5th day (both bowl 40 balls)	**1**

Total marks: 10

Question 11

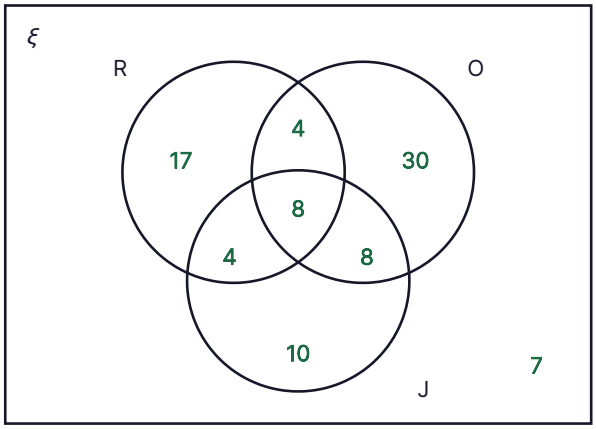
(i)	
$\angle OAP = 90°$ and $\angle OBP = 90°$ (a tangent is perpendicular to the radius at the point of contact)	**1**
$\angle OAP + \angle OBP = 90° + 90° = 180°$	**1**
Opposite angles of APBO are supplementary, so APBO is a cyclic quadrilateral	**1**
(ii)	
$\angle AOB = 2 \times \angle ACB$ (angle at the centre is twice the angle at the circumference, on the same arc AB)	**1**
$\angle APB + \angle AOB = 180°$ (opposite angles of the cyclic quadrilateral APBO)	**1**
$\angle APB = 180° - 2 \times \angle ACB$	**1**
(iii)	
$\angle ACB$	**1**
Reason: angle in the alternate segment (tangent–chord angle theorem: the angle between the tangent PA and the chord AB equals the angle in the alternate segment)	**1**
$\angle PBA$	**1**
Reason: PA = PB (tangents from an external point are equal), so in the isosceles triangle PAB the base angles are equal	**1**

Total marks: 10

Question 12

(i)	
Venn diagram copied with the sets R, O, J and the universal set; 7 entered outside the circles	**1**
Customers who bought at least one kind = 88 − 7 = 81	**1**
Two or more kinds = 81 − 57 = 24	**1**
(ii)	
Let all three = g; exactly two kinds = $2g$	**1**
$g + 2g = 24$	**1**
$g = 8$ (and 16 bought exactly two kinds)	**1**
(iii)	
Only R and O = 4, so (only R and J) + (only O and J) = 16 − 4 = 12	**1**
$e + 2e = 12$: only R and J = 4; only O and J = 8	**1**
(iv)	
Only R = 33 − (4 + 4 + 8) = 17; only J = 30 − (4 + 8 + 8) = 10	**1**
Only O = 57 − 17 − 10 = 30	**1**

Total marks: 10



Completed Venn diagram